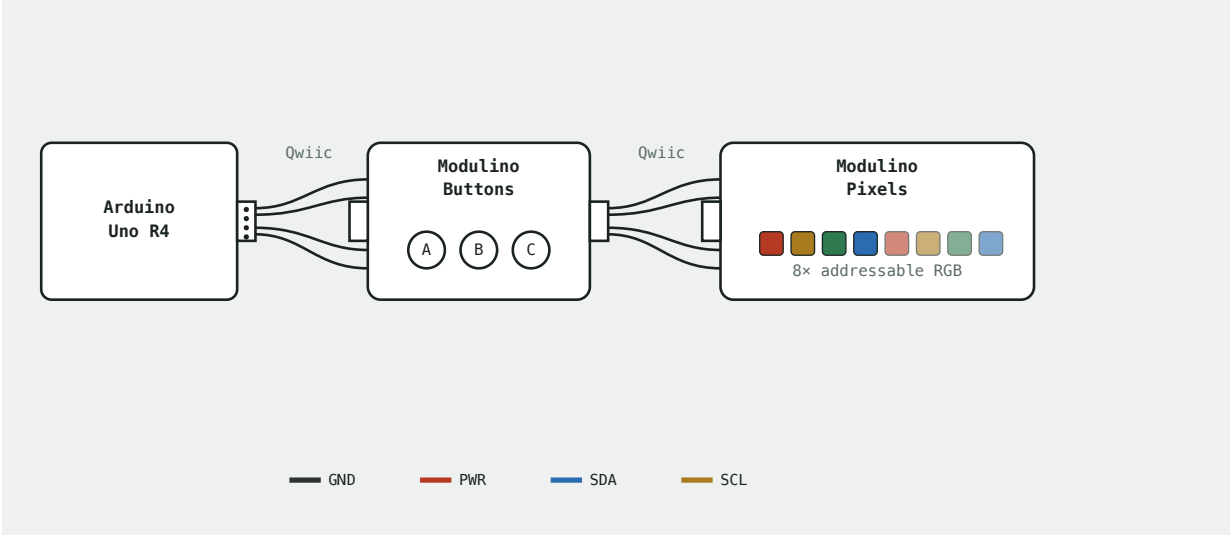
Traffic light simulator

A push-button traffic light using two Modulino modules and no other wiring. Press A for green (go), B for red (stop) or C for yellow (caution); the pressed button's own LED lights up and the whole Pixels strip fills with the matching color.

Board · Arduino Uno R4 **Modules** · Modulino Buttons, Modulino Pixels
New concepts · digital input, `if` / `else if`



Both modules chain on the Qwiic bus. There's only one of each type in the kit, so no addressing is needed and the order doesn't matter.

PIN & CONNECTION REFERENCE

Component	Connects to
Modulino Buttons	Qwiic/I2C bus (Arduino's Qwiic connector)
Modulino Pixels	Qwiic/I2C bus (chained from the Buttons module)

REQUIRED LIBRARIES

- `Arduino_Modulino` — pulls in the sensor-vendor dependencies automatically

CODE REFERENCE

Reading which button was pressed

`update()` returns true only when something changed, so repeat presses don’t flood your logic.

```
// --- Inside loop() ---
if (buttons.update()) {
  if (buttons.isPressed('A')) {
    // ...
  } else if (buttons.isPressed('B')) {
    // ...
  }
}
```

Filling the whole strip

Nothing changes on the strip until you call `show()`.

```
// --- Helper function (below loop()) ---
void showTrafficLight(ModulinoColor color)
{
  for (int i = 0; i < 8; i++) {
    pixels.set(i, color, 40);
  }
  pixels.show();
}
```

Lighting the button’s own LED

One call sets all three button LEDs.

```
// --- Inside loop() ---
buttons.setLeds(true, false, false);
// A on, B off, C off
```

TRY IT YOURSELF

- Hold all three buttons at once for a flashing-amber mode.
- Light only the pixels nearest each button’s own position instead of filling the whole strip one color.